

Dijkstra's Algorithm on Indian Cities Road Network

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1. Introduction

This project implements Dijkstra's Algorithm (Uniform Cost Search) to compute the shortest path between cities using real-world road distance data from a CSV file.

2. Algorithm Explanation

Dijkstra's Algorithm is a greedy algorithm that finds the shortest path from a source node to all other nodes in a weighted graph. It always expands the node with the minimum distance.

3. Requirements

- C Compiler (GCC)
- CSV file containing city connections

4. CSV Format

```
Origin, Destination, Distance
Agra, Delhi, 240
Agra, Lucknow, 334
Ahmedabad, Mumbai, 526
```

5. How to Run the Project

```
gcc dijkstra_csv.c -o dijkstra
./dijkstra
```

6. Output

```
Shortest distances from Agra:
Delhi -> 240 km
Lucknow -> 334 km
Kanpur -> 277 km
```

7. Measures of Effectiveness

- Path Length
- Nodes Explored
- Execution Time
- Accuracy

8. Applications

- GPS Navigation Systems
- Google Maps Routing
- Robotics Path Planning
- Logistics Optimization

9. Conclusion

This project demonstrates how graph search algorithms like Dijkstra's Algorithm are applied in real-world navigation systems.